

Yash Totlani

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Mathematics and Physics undergraduate with hands-on experience building AI/ML, LLM, and multi-agent systems across finance, cloud infrastructure, and scientific computing. Combines first-principles mathematical rigor with practical engineering execution, including agentic financial workflows using CrewAI, MCP, and the OpenAI Agents SDK, stress-testing models at Goldman Sachs, LLM pipelines at a fintech startup, deep learning research grounded in optimal transport theory, and large-scale simulations of quantum Hamiltonians.

EDUCATION

University of Texas at Austin

Austin, TX

Bachelor's of Science Honors in Mathematics and Physics

Aug 2022 – May 2026

- **Graduate Coursework:** Advanced Algebraic Structures, Solid State Physics, Algebraic Geometry, Algebraic Number Theory, Representation Theory, Algebraic Topology, Riemann Surfaces
- **Undergraduate Coursework:** Vector Calculus, Linear Algebra, Data Structures & Algorithms, Differential Equations, Real Analysis, Modern Physics, Electrodynamics

PROFESSIONAL EXPERIENCE

Software Engineering Summer Analyst

June 2024 – Aug 2024

Goldman Sachs

Dallas, TX

- Supported funding decisions across business units by monitoring liquidity and balance sheet risk in corporate treasury, analyzing \$10B+ in exposures
- Enhanced net interest income forecasting models by upgrading division-wide systems to meet regulatory standards, improving accuracy by 15%+ and reducing reporting discrepancies by 20% using Python, SQL, and Pandas-based validation checks
- Built quantitative modeling tools using Python, NumPy, and SciPy to analyze stress test outputs across 100+ scenarios, identifying key drivers of balance sheet degradation under adverse macroeconomic conditions
- Automated data flow logging in the stress testing pipeline with Python and SQL, reducing manual reporting overhead by 60%+ and improving auditability and reliability of regulatory submissions

Machine Learning Engineer

May 2025 – Aug 2025

hScore Advisors

San Francisco, CA

- Developed a GenAI-powered financial query agent using Python, LangChain, GraphQL, and agentic retrieval workflows to process natural-language financial queries, improving data mapping accuracy by 25%.
- Implemented retrieval and ranking logic using embeddings, Pandas-based feature processing, and structured GraphQL queries to map free-text financial queries to relevant entities
- Integrated real-time and historical market data, reducing manual analyst input by 40%
- Owned end-to-end pipeline from prototype to deployment, contributing core features to hScore's public beta

Postman Agentic HR Project

May 2026

Postman, Recruiting Automation Demo

San Francisco, CA

- Built a CrewAI-based recruiting workflow with specialized sourcing and screening agents to identify public candidate leads and evaluate role fit for a Postman job requisition
- Integrated SerperDevTool-powered search to retrieve evidence-backed candidate profiles, capturing source URLs, matched job-relevant terms, uncertainty notes, and sourcing audit metadata
- Generated structured JSON outputs and a self-contained HTML recruiter dashboard summarizing shortlisted candidates, value rationale, requirement matches, and human-review flags

Machine Learning Engineer

June 2023 – Aug 2023

XTaaS Cloud

San Francisco, CA

- Built Python and SQL-based data pipelines to evaluate voice-to-text system performance across 50K+ transcripts, using Pandas/NumPy for analysis and improving efficiency by 40%
- Developed a custom text transcription solution deployed via Docker, reducing operating costs by 25%+ through optimized processing workflows
- Implemented an NLP classification model using Python and TensorFlow to label call transcripts as successful or unsuccessful connections, achieving 90%+ accuracy across large-scale datasets

RESEARCH EXPERIENCE

Computational Condensed Matter Researcher

January 2025 – September 2025

University of Texas at Austin

Austin, TX

- Worked with Dr. Junyeong Ahn on electronic structure in topological quantum materials
- Applied analytical methods and large-scale numerical analysis using Python, NumPy, and SciPy to extract structural patterns in high-dimensional mathematical models
- Developed mathematical and computational models to analyze band structure dynamics of materials under strong periodic (AC) driving fields
- Implemented numerical simulations of quantum Hamiltonians in Python/C++ to study how external signals modify system spectra and emergent structure and visualized spectral evolution with Matplotlib

Machine Learning Researcher

June 2020 – June 2023

University of Texas at Austin

Austin, TX

- Conducted research with faculty on machine learning models based on dynamic optimal transport and continuous-time dynamical systems, developing deep learning architectures in TensorFlow/PyTorch that model neural networks as continuous-time flows
- Built numerical experiments using Python, NumPy, SciPy, and Matplotlib to study stability, convergence behavior, and training dynamics of continuous-time models

PROJECTS

AI Recruiting Value Screening System | *Python, CrewAI, SerperDevTool, JSON, HTML, Cursor*

- Built a multi-agent recruiting workflow using CrewAI to coordinate a talent sourcing agent and candidate screening agent for role-specific candidate discovery and evaluation.
- Integrated SerperDevTool to execute approved public search queries and capture evidence-backed candidate leads with source URLs, matched terms, uncertainty notes, and sourcing audit metadata.
- Designed a value-based screening pipeline that compares candidates against job requirements, role value drivers, and market-rate context to generate directional expected annual business value estimates.
- Generated structured JSON outputs and a self-contained HTML recruiter dashboard summarizing screened candidates, shortlist decisions, human-review flags, and rationale for each candidate.

Multi-Agent Financial Research System | *Python, CrewAI, OpenAI Agents SDK, MCP, Cursor*

- Built a multi-agent financial analysis system using CrewAI to coordinate specialized agents for market research, risk analysis, data retrieval, and investment memo generation.
- Implemented MCP-based tool integrations to expose structured financial data, portfolio files, and external APIs to LLM agents through typed tool interfaces.
- Used the OpenAI Agents SDK to prototype agent handoffs, tool-calling workflows, and traceable execution paths for multi-step financial reasoning tasks.
- Designed an evaluation workflow comparing agent outputs against baseline analyst templates, improving consistency of generated financial summaries and reducing manual research effort.

AI Codebase Onboarding Agent | *Python, MCP, OpenAI Agents SDK, CrewAI, Cursor*

- Built an AI codebase assistant that answers natural-language questions about unfamiliar repositories by retrieving and reasoning over relevant source files.
- Created an MCP server exposing typed tools for file search, source-code retrieval, dependency inspection, and documentation lookup.
- Used the OpenAI Agents SDK to coordinate tool calls, maintain execution traces, and generate grounded explanations of project architecture and implementation details.
- Implemented a CrewAI workflow with separate agents for code summarization, documentation generation, test review, and refactoring recommendations.

Light-Induced Current Simulation in Graphene | *Python, Mathematica, Numerical Modeling, Scientific Computing*

- Developed a computational model to study how circularly polarized light can alter electron behavior in graphene and generate Hall-like current flow without a magnetic field.
- Simulated a graphene ribbon under different light intensities and frequencies to analyze changes in energy structure and transport behavior.
- Implemented matrix-based numerical methods to compute current response under small applied voltages and compare behavior across simulation parameters.

- Created visualizations of energy spectra and current trends to communicate complex quantum transport results in a clear, interpretable format.

Monte Carlo Tree Search + REINFORCE | *Python*

- Derived and implemented the REINFORCE gradient estimator from the policy gradient theorem
- Designed a custom hybrid objective combining Monte Carlo value estimates with policy-gradient updates
- Analyzed variance reduction behavior empirically using NumPy for experiments and Matplotlib for diagnostics

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, R, JavaScript

AI/ML: PyTorch, TensorFlow, LangChain, CrewAI, OpenAI Agents SDK, MCP, embeddings, RAG

Data/Scientific Computing: Pandas, NumPy, SciPy, Matplotlib

Developer Tools: Git, Docker, Cursor, AWS, Azure

Mathematical Tools: Linear Algebra, Stochastic Calculus, Functional Analysis, Group Theory, Spectral Theory, Differential Geometry, Number Theory

CERTIFICATIONS

- AWS Cloud Practitioner
- Google ML Engineer
- Microsoft Azure Fundamentals